

Virtual PwC Switzerland Internship Report

Introduction

This document outlines the key tasks and deliverables from my virtual internship with PwC Switzerland. It simulated a company, PhoneNow, a leading call center and telecommunications provider. The primary objective was to leverage data analysis and visualization to create three strategic dashboards: one for call center operations, one for customer retention, and another for HR diversity, providing actionable insights to key stakeholders.

All tasks were performed with Microsoft Power BI.

Task 1: PhoneNow Call Centre Dashboard

Background Information

- **Company/Department:** PhoneNow Call Centre Operations
- **Project Goal:** To create a real-time performance dashboard for a call center manager to monitor and optimize daily operations and agent productivity. The goal is to provide a single view to quickly identify operational bottlenecks and performance trends.
- **Key Context:** Call centers rely on key metrics to ensure high-quality customer service and efficient operations. A manager needs to be able to see an overview of call volumes, agent performance, and customer satisfaction at a glance to make informed decisions and manage staffing levels.

Task Performed

- **My Role:** Data Analyst / Dashboard Developer
- **Specific Actions:** I designed and outlined a dashboard based on the provided KPIs.
 - **Overall Customer Satisfaction:** A single-value card or a gauge to show the average satisfaction score, providing an immediate health check of customer sentiment.
 - **Overall Calls Answered/Abandoned:** A stacked bar chart or a pie chart to visualize the ratio of answered to abandoned calls, a critical metric for understanding customer service accessibility.

- **Calls by Time:** A time series chart (line chart) showing call volume over the course of a day, a week, or a month to identify peak hours and inform staffing schedules.
- **Average Speed of Answer:** A single-value card showing the average time it takes for an agent to answer a call, with a color-coded alert system for poor performance.
- **Agent's Performance Quadrant:** A scatter plot or quadrant chart to analyze individual agent performance by plotting **Average Handle Time vs. Calls Answered**, which helps the manager identify top performers, those needing support, or those with potential issues.

Outcome/Deliverables

- **Deliverable:** A comprehensive dashboard design and a list of visuals tailored to meet the operational needs of the call center manager.
- **Results:** This design provides a clear, actionable tool for call center management to improve efficiency, reduce abandoned calls, and boost overall customer satisfaction.

Task 2: PhoneNow Customer Retention Dashboard

Background Information

- **Company/Department:** PhoneNow Marketing & Customer Retention
- **Project Goal:** To create an analytics dashboard focused on customer churn and retention. The goal is to identify at-risk customers, understand churn drivers, and inform targeted marketing campaigns to retain the most valuable customers.
- **Key Context:** Customer churn is a major challenge for telecommunications companies. Understanding who is leaving, why they are leaving, and which services they use is crucial for developing effective retention strategies. This dashboard provides a 360-degree view of the customer base.

Task Performed

- **My Role:** Data Analyst / Dashboard Developer
- **Specific Actions:** I planned a dashboard to visualize key customer data points.
 - **Customers who left within the last month:** A single-value card with a month-over-month trend line to track recent churn rates.
 - **Services each customer has signed up for:** A bar chart showing the percentage of

customers subscribed to each service (e.g., streaming, online security) to identify which services are popular and which may be underutilized by those who have churned.

- **Customer account information:** A detailed table or filtered view for analysts to review key metrics for specific customers, such as **how long they have been a customer, their contract type, and monthly charges**.
- **Demographic info about customers:** Bar charts or histograms for **gender, age range, partner, and dependents** to identify demographic trends among different customer segments and those who have left the service.

Outcome/Deliverables

- **Deliverable:** A customer retention dashboard design that provides deep insights into the customer base and identifies key factors contributing to churn.
- **Results:** This dashboard will enable the marketing and retention teams to make data-driven decisions, leading to more effective campaigns and a reduction in customer churn.

Task 3: HR Diversity Analysis

Background Information

- **Company/Department:** PhoneNow Human Resources
- **Project Goal:** To analyze workforce diversity and gender pay equity within the organization. This analysis will provide HR leadership with insights to promote a more inclusive and equitable workplace.
- **Key Context:** In today's corporate environment, companies are increasingly focused on diversity, equity, and inclusion (DEI). The HR department at PhoneNow needs to understand the current state of its workforce across various departments to identify any existing imbalances or disparities in pay and representation.

Task Performed

- **My Role:** Data Analyst / People Analytics
- **Specific Actions:** I performed an HR diversity analysis on the PhoneNow employee dataset, focusing on two key areas.

- **Gender Diversity:** I analyzed the distribution of male and female employees across different departments to identify any underrepresented groups. I also created a visual to compare the average salary between genders.
- **Diversity by Department:** I broke down the gender and salary data by department to highlight specific areas that may require targeted hiring or salary adjustments to promote better equity. The analysis also included the number of tickets opened in the categories administrative and technical.

Outcome/Deliverables

- **Deliverable:** A comprehensive HR diversity analysis report and an interactive dashboard.
- **Results:** The analysis provides the HR department with a clear picture of workforce diversity, enabling them to make data-driven decisions to foster a more equitable and inclusive environment.